Anticipatory Routing

Route step t with the parameters of step t–Δt

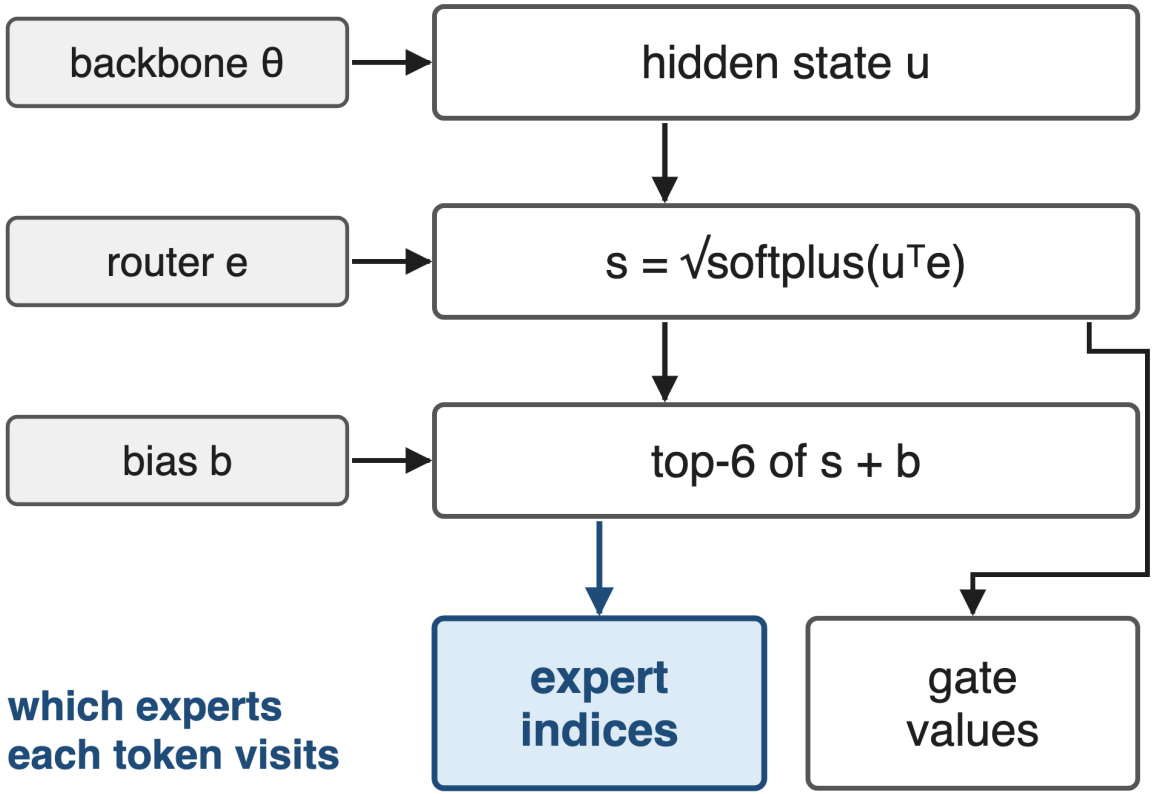
DeepSeek-AI · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

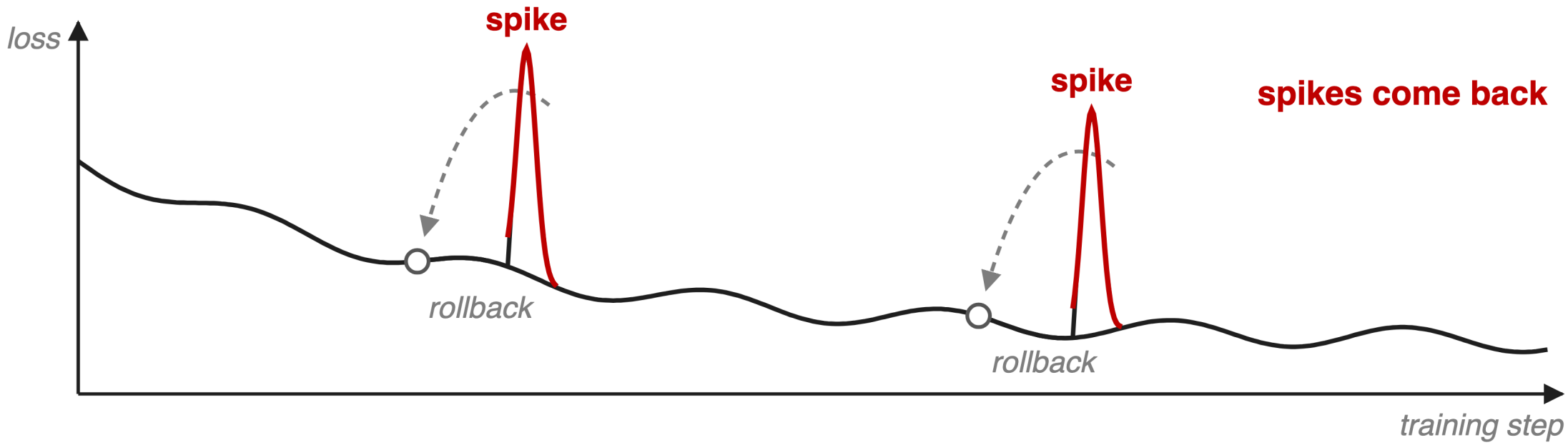
Optimizer & Stability · arxiv.org/abs/2606.19348

The router picks experts from current parameters

- V4-Pro: **1.6T** params, **384** experts, top-**6**
- **33T** tokens, batches up to **94.4M** tokens
- Expert choice depends on backbone, router and bias



Rollbacks restore the run, not the stability

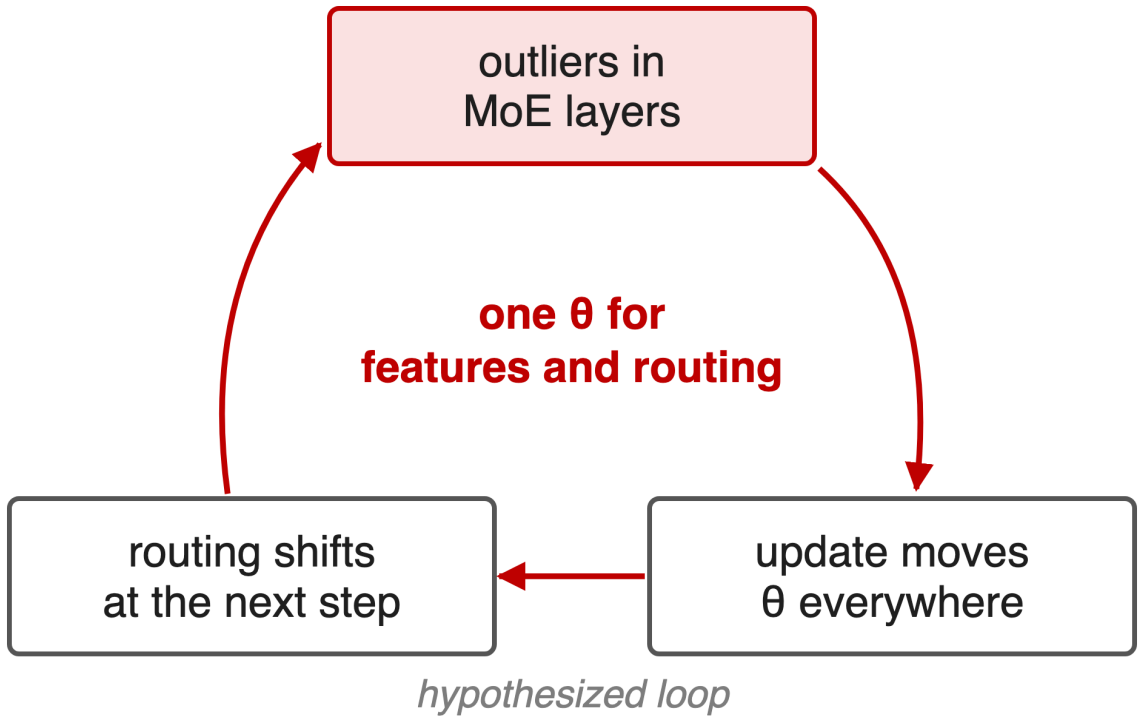


Schematic; DeepSeek publishes no spike sizes, counts or lost compute

Spikes track outliers in MoE layers, and routing appears to amplify them

Routing and features update in lockstep

- Standard step: one θ(t) for features and routing
- An update that creates outliers also re-routes tokens



DeepSeek calls the exact mechanism an open question

Insight: let routing lag the backbone by Δt

- Empirical: decoupling backbone and router updates
- Training becomes markedly more stable

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Compute with θ(t), but choose experts with θ(t–Δt)

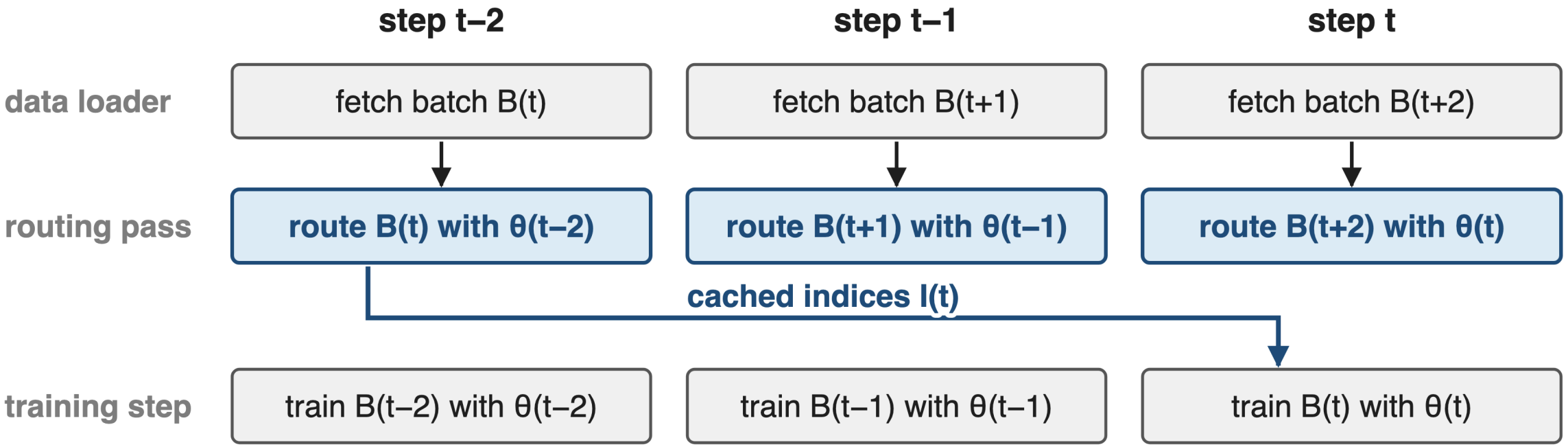
Step t computes with θ(t), routes with θ(t–Δt)

	(a) Standard step t	(b) Anticipatory step t
which experts (top-6 indices)	θ(t)	**θ(t–Δt) · cached**
hidden states and expert outputs	θ(t)	θ(t)
loss and gradients	θ(t)	θ(t)
optimizer update	θ(t) → θ(t+1)	θ(t) → θ(t+1)

Gate values (expert weights): version not stated in the paper

Only one row changes: which experts each token visits

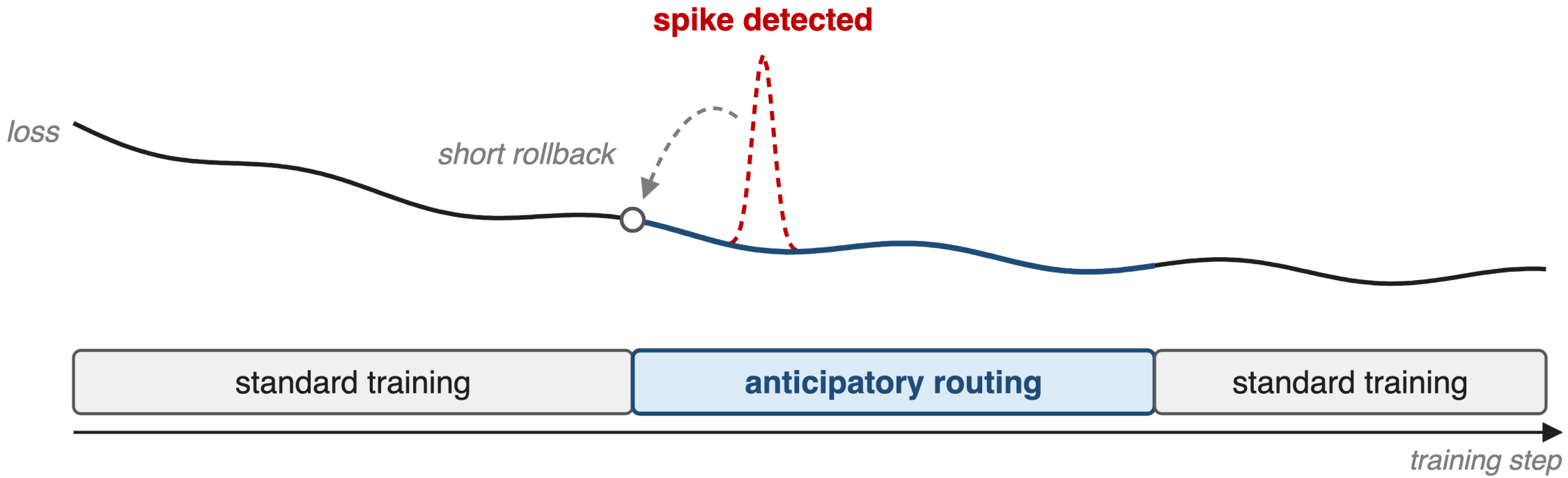
❶ Routing indices are computed Δt steps early



each routing pass feeds the step Δt later (Δt = 2 here)

θ(t–Δt) is already in memory at step t–Δt: no second copy of the model

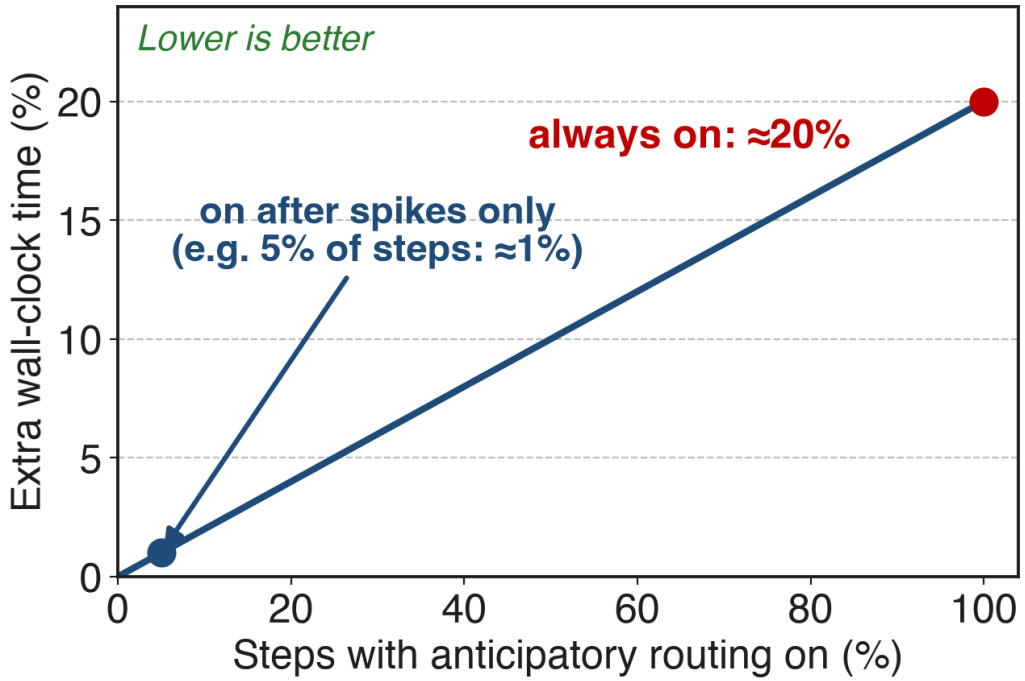
❷ Anticipatory mode turns on only after a spike



Detect a spike, roll back briefly, route anticipatorily for a while, then return

≈20% slower while on, near zero overall

- About **20%** extra wall-clock while active
- Active only after detected spikes
- DeepSeek: spikes averted, quality unchanged

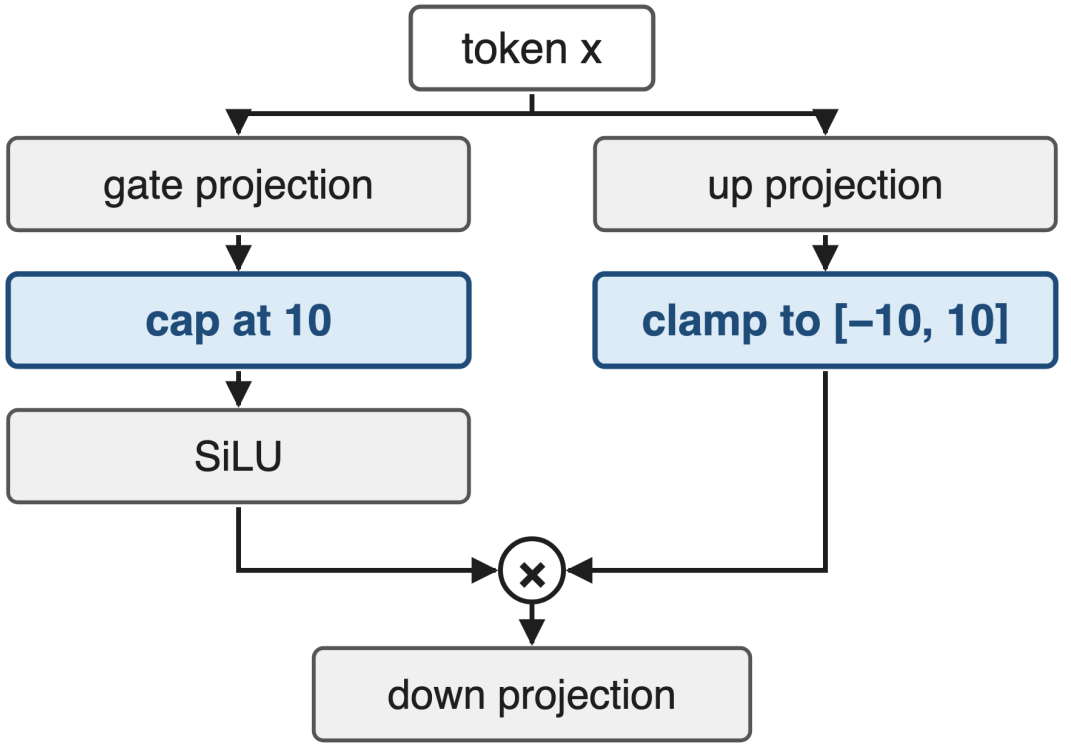


Analytic: overhead ≈ 20% × active fraction; the 5% point is illustrative

Pay for the lookahead only while a spike is being handled

V4 pairs it with SwiGLU clamping

- Clamping removes outliers directly
- Linear part clamped to **[–10, 10]**, gate capped at **10**
- Used throughout V4-Flash and V4-Pro training



The cost: an extra pass, rollbacks, no theory

- **≈20%** extra wall-clock while active
- Prefetch batches Δt steps ahead, cache indices
- Needs a spike detector and rollbacks
- Δt and thresholds unpublished
- No ablations or spike statistics
- Why it works: still open

Summary

- V4 loss spikes recurred after plain rollbacks
- **Insight**: route with $\theta(t-\Delta t)$, compute with $\theta(t)$
- One cached routing pass, on only after spikes
- **≈20%** while on, **negligible** overall

https://arxiv.org/abs/2606.19348